

Section - B

_____ Q:2(i) _____
 ← _____ A:2(i) _____

Five basic properties of real numbers :-

- 1) Closure property :- all $a, b \in \mathbb{R}$, $a+b \in \mathbb{R}$
- 2) Associative property :- $(a+b)+c = a+(b+c)$
- 3) Commutative property :- $a+b = b+a$
- 4) Additive property :- $a+0 = a$, $0+a = a$
- 5) Additive inverse property :- $(a)-a = 0$

_____ Q:2(ii) _____
 _____ A:2(ii) _____

a) $5 \times 7 = 7 \times 5$
 Commutative property

b) $3+(4+2) = (3+4)+2$
 Associative property

c) $8 \times 1 = 8$
 Multiplicative identity.

Day: _____

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Q: 2 (iii) (or)

A: 2 (iii) (or)

a) $\sqrt[3]{6} \div \sqrt{5}$

Solution:-

$$= \sqrt[3]{6 \times 5}$$

$$= \sqrt[3]{30}$$

~~$$= \sqrt[3]{30} \times 3$$~~

∞

b) $\sqrt{20} \div \sqrt{5}$

Solution:-

$$= \sqrt{\frac{20}{5}}$$

$$= \sqrt{4}$$

$$= 2$$

Q: 3 (iv) (or)

A: 3 (iv) (or)

a) $x^{\frac{3}{4}}$

Sol:-

$$= x^{\frac{3}{4}}$$

$$= (x^3)^{\frac{1}{4}}$$

$$= \sqrt[4]{x^3}$$

Q:2(v)
A:2(v)

a) $\sqrt{3n+1} = 5$

Taking squares on both sides:-

$$(\sqrt{3n+1})^2 = (5)^2$$

$$\Rightarrow 3n+1 = 25$$

$$\Rightarrow 3n = 25-1$$

$$\Rightarrow 3n = 24$$

$$\Rightarrow n = \frac{24}{3}$$

$$= 8$$

Check :-

$$\Rightarrow \sqrt{3(8)+1} = 5$$

$$\Rightarrow \sqrt{24+1} = 5$$

$$\Rightarrow \sqrt{25} = 5$$

$$\Rightarrow 5 = 5$$

Solution set $\{8\}$

Q:2(vi)
A:2(vi)

a) 7340,000

Solution :- 7.34×10^6

b) 0.000089

8.9×10^{-5}

c) 125.6 : 1.256×10^2

Day: _____

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Q: 2 (vii) (or)A: 2 (vii) (or)

$$a) \log_3 x = 4$$

$$\Rightarrow x = (3)^4$$

$$x = 81$$

$$b) \log_x 16 = 2$$

$$\Rightarrow 16 = (x)^2$$

$$\Rightarrow (4)^2 = (x)^2$$

$$4 = x$$

$$c) \log_4 64 = x$$

$$\Rightarrow 64 = (4)^x$$

$$\Rightarrow (4)^3 = (4)^x$$

$$3 = x$$

Q: 2 (viii) (or)A: 2 (viii) (or)

Common logarithm	Natural logarithm
Base	
* It is a rational number such as (10)	It is supposed to be an irrational number such as (e)
Notation	
$\log_{10} x$	$\log_e x$

Made by

* By Henry Briggs By John Napier

$$\frac{\text{Q: 2 (ix) (or)}}{\text{A: 2 (ix) (or)}}$$

a) $\log 67.3$

$$\log = 67.3$$

$$= 6.73 \times 10^1$$

Characteristic :- 1

b) $\log 0.00892$

$$\log = 0.00892$$

$$= 8.92 \times 10^{-3}$$

Characteristic :- $\bar{3}$

c) $\log 5$

$$\log = 5.0 \times 10^0$$

Characteristics :- 0

$$\frac{\text{Q: 2 (u) (or)}}{\text{A: 2 (u) (or)}}$$

a) $2 \log x - 3 \log y + \log z$

Sol:-

$$\log x^2 - \log y^3 + \log z$$

$$\log x^2 \times + \log z$$

$$\log \frac{x^2}{y^3} + \log z$$

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$$\log\left(\frac{x^2 z}{y^3}\right) \quad \text{Solution}$$

$$\begin{array}{l} \text{Q: } z(x) \\ \text{A: } z(x) \end{array}$$

a) 2^{20}

$$\Rightarrow \log 2 \times 20$$

$$\Rightarrow 0.3010 \times 20$$

$$\Rightarrow 6.02$$

Char :- 6

$$\begin{aligned} \text{Characteristic} + 1 &= 6 + 1 \\ &= 7 \text{ digits.} \end{aligned}$$

Section - C

Q: 3(i)(or)

Q: 3(i)(or)

Solution:-

$$= \left(\frac{x^p}{x^q} \right)^{p+q} \times \left(\frac{x^q}{x^r} \right)^{q+r} \times \left(\frac{x^r}{x^p} \right)^{r+p} = 1$$

$$= (x^{p-q})^{p+q} \times (x^{q-r})^{q+r} \times (x^{r-p})^{r+p} = 1$$

$$= x^{p^2-q^2} \times x^{q^2-r^2} \times x^{r^2-p^2} = 1$$

$$= x^{p^2} \times x^{-q^2} = 1 \quad \dots$$

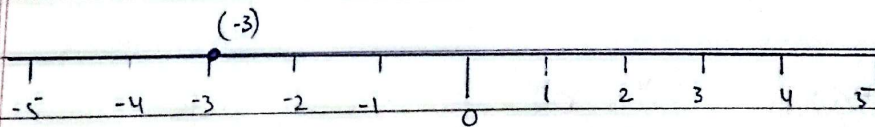
$$1 = 1 \quad \dots$$

Q: 3(ii)

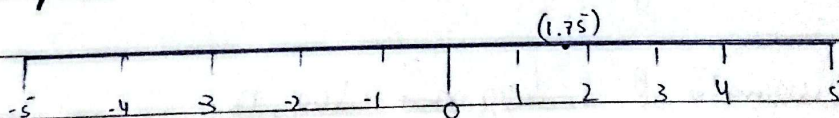
A: 3(ii)

Solution

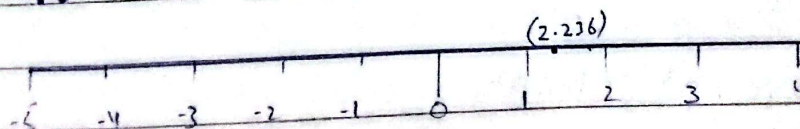
a) -3



b) $7/4 = 1.75$



c) $\sqrt{5} = 2.236$



d) $-2 < x < 3$



ii) $(x+y)+z = x+(y+z)$
Associative property

iii) $5 \times \left(\frac{1}{5}\right) = 1$

Multiplicative inverse property

Q:3(iii)
A:3(iii)

a) let $\log_b m = x$ and $\log_b n = y$

$\log_b m = x$, $\log_b n = y$

$m = b^x$ $n = b^y$

$m \times n = b^x \times b^y$

$mn = b^{x+y}$

$mn = x+y$

$\log_b(mn) = \log_b m + \log_b n$

b) 72

$72 = 2^3 \times 3^2$

$= \log 2^3 \times \log 3^2$

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$$= 3 \log 2 + 2 \log 3$$

$$= 3(0.3010) + 2(0.4771)$$

$$= 0.9030 + 0.9542$$

$$= 1.8572 \text{ Answer}$$

Q: 3(iv)(or)

A: 3(iv)(or)

$$= \text{Numbers of digits} = \text{char of log} + 1$$

$$5^{37}$$

$$= \log 5 \times 37$$

$$0.6990 \times 37$$

$$= 25.863$$

$$\text{Characteristic} = 25$$

$$= 25 + 1$$

$$= 26 \text{ digits.}$$

Day: _____

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MCQs :-

- 1) C ($\sqrt{5}$)
- 2) C (Distributive)
- 3) C (rationals and irrational)
- 4) D (7)
- 5) C (radicand)
- 6) B (1)
- 7) A (5.2×10^{-5})
- 8) C (6)

- 9) C (2)
- 10) A ($\log m + \log n$)
- 11) D (e)
- 12) A (0)